The Effects of Patient and Hospital Factors on Hospital-Acquired Infection Risk.

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1 Overview

Hospital-acquired infections (HAIs), or healthcare-associated infections, are a major problem in United States hospital systems. Hospital-acquired infections are infections that a patient contracts while staying at the hospital. HAIs are linked to many factors, such as the patient's age or length of stay at a hospital, as well as the the hospital's infrastructure, such as the number of available hospital beds and hospital staff. Knowing how these factors affect a patients' risk of acquiring hospital-acquired infections is important for hospital leaders and systems to implement policies in minimizing the risk of hospital-acquired infections and better patient care.

The main research question we will investigate is: How do patient and hospital factors affect the risk of a hospital-acquired infection in a hospital?

We define what "Patient" and "Hospital" factors are in section two.

2 Data Description and Processing

The dataset contains data collected from hospitals in the United States, with each row representing one hospital. The dataset contains 113 rows (observations) and 11 columns (variables).

We categorized the variables in the data set in terms of "Patient Factors" and "Hospital Factors". "Patient Factors" are variables that reference information about a patient, such as how long they stay at the hospital, and their age. "Hospital Factors" are variables associated with the hospital, such as the number of beds the hospital has and the number of nurses for staff. Additionally, we placed "Culture" and "Xray" variables under the group "Hospital Factors", as these are tests conducted by hospitals on patients.

Patient Factors:

Stay (numeric): average length of patient stay (days)

Age (numeric): average patient age (years)

Hospital Factors:

Culture (numeric): Ratio of number of cultures (testing a patient's sample for infection-causing microorganisms) performed to number of patients without signs or symptoms of hospital-acquired infection, times 100.

Xray (numeric): Ratio of number of X-rays (taking an image of the body, particularly the chest, to look for conditions such as pneumonia) performed to number of patients without signs or symptoms of pneumonia, times 100.

Beds (numeric): Average number of beds in hospital during study period.

MedSchool (categorical - factor): Hospital has an associated medical school, where:
–> 1 = Yes
–> 2 = No

We decided to factor this categorical variable, MedSchool, as there are different "groups" or levels for medical school. For example, "1" represents "Yes" as schools with a medical school, and "2" represents "No" as schools without a medical school.

Region (categorical - factor): Geographic region, where:
–> 1 = NE
–> 2 = NC
–> 3 = S
–> 4 = W

We decided to factor this categorical variable, Region, as there are different "groups" of regions. One group of hospitals are located in the Northeast, and three distinct groups are located in the distinct regions of the North Central, South, and West in the United States.

Census (numeric): Average number of patients in hospital per day during study period.

Nurses (numeric): Average number of full-time equivalent registered and licensed practical nurses during study period (number full-time plus one half the number part time).

Facilities (numeric): Percent of 35 potential facilities and services that are provided by the hospital.

Outcome/ Dependent Variable: Our dependent variable is infection risk (named InfctRsk in our dataset). InfctRsk (numeric): Average estimated probability of acquiring infection in hospital (percent)

The data set was checked for missing values. Of the original 116 observations, we removed 3 observations that contained missing values leaving 113 complete observations.

We used the InfctRsk (infection risk) variable as our outcome.

We did not find or remove any outliers in our dataset.

3 Data Exploration

Our first variable to explore is our outcome variable infection risk, labeled InfctRsk in our dataset. This numeric variable is the average estimated probability of acquiring infection in a hospital, written as a percent.

According to the descriptive statistics table "Infection Risk (InfctRsk) Descriptive Statistics", the minimum average infection risk is 1.3% and the average maximum infection risk is 7.8%. The data centers around a mean of 4.35% and the median average infection risk is 4.4%.

Table 1: Infection Risk (InfctRsk) Descriptive Statistics

statistic	value
Min	1.3
Mean	4.35
Median	4.4
Max	7.8
StdDev	1.34

To further explore the infection risk (InfctRsk) variable, a histogram showing the distribution of the variable is constructed. This displays an approximately normal distribution, with one peak.

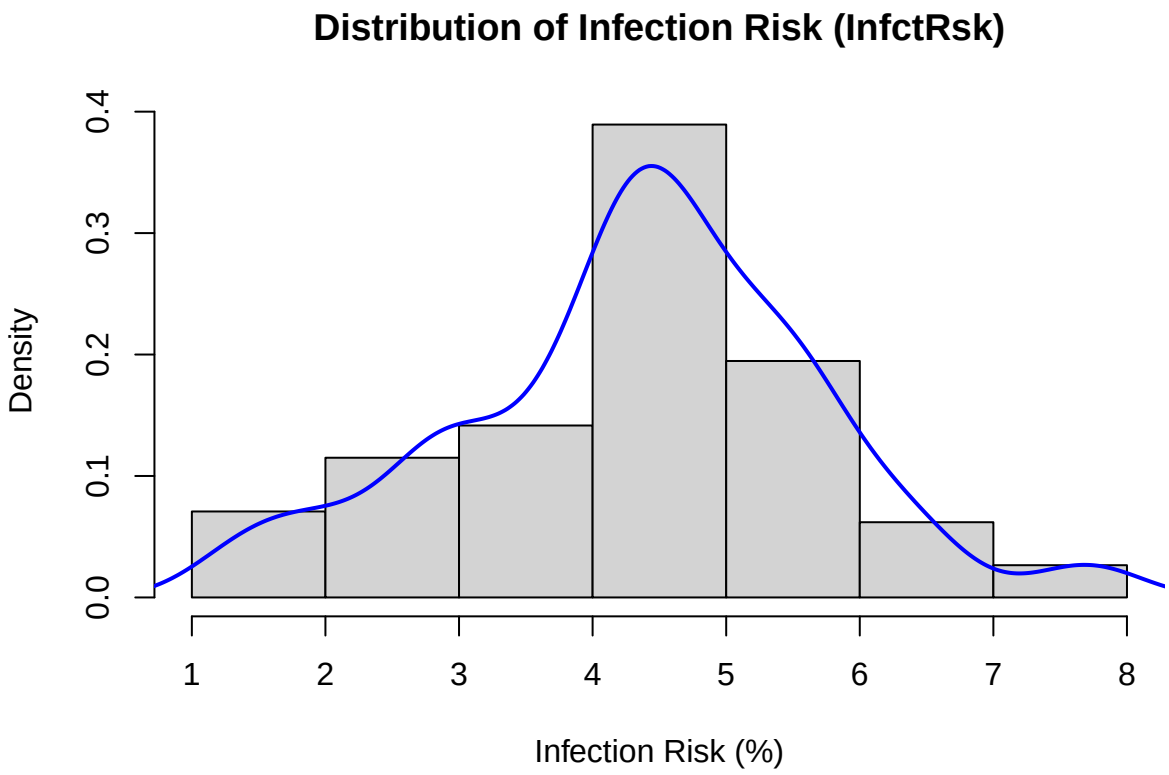


Figure 1: Histogram showing distribution of InfctRsk Variable, with Density Curve Overlaid

The density curve fits the data pretty well.

The descriptive statistics table matches up with the histogram as we can see that there is a minimum somewhere between 1% and 2% (Actual minimum: 1.3%), a maximum is located between 7% and 8% (actual: 7.8%), and the average/mean for the InfctRsk variable is between 4% and 5% (Actual average or mean: 4.35%).

3.1 Patient Factors

The two "Patient" factors in our dataset are Stay and Age. We explored the distributions of these variables below in sections 3.1.1 and 3.1.2.

3.1.1 The Variable Stay

We start with the variable Stay, which measures average length of patient stay, in days, at a hospital.

The table "Stay Descriptive Statistics" for the variable Stay tell us that the minimum average length that patients stayed at a hospital is 6.7 days, and the maximum average length that patients stayed at a hospital is 19.56 days. The largest cluster of hospitals have patients stay for an average length of 9.65 days.

The histogram's distribution is skewed to the right, with the average length of Stay across most hospitals is less than two weeks. The largest cluster of hospitals maintains an average patient stay of 8 to 10 days. Based on the skewness and outliers, we can see that there are few hospitals that have much longer average stays, ranging from 16 to 20 days.

The density curve fits the histogram's distribution pretty well.

Table 2: Stay Descriptive Statistics

statistic	value
Min	6.7
Mean	9.65
Median	9.42
Max	19.56
StdDev	1.91

3.1.2 The Variable Age

The variable Age measures the average patient age is in years.

The histogram for the variable Age reveals an approximately normal distribution, with one peak between 50 and 55 years. This shape is supported by the corresponding descriptive statistics table "Age Descriptive Statistics", which prove that the mean average patient age is 53.23 years old. The minimum average patient age is in the range of 35 and 40 years, and the corresponding descriptive statistics reveal an actual minimum of 38.8 years. Additionally, the histogram reveals a maximum average age at one hospital is in the range of 65 and 70 years, with the descriptive statistics revealing the actual maximum average age to be 65.9 years old

The density curve fits the histogram's distribution well, confirming a normal distribution of the variable Age.

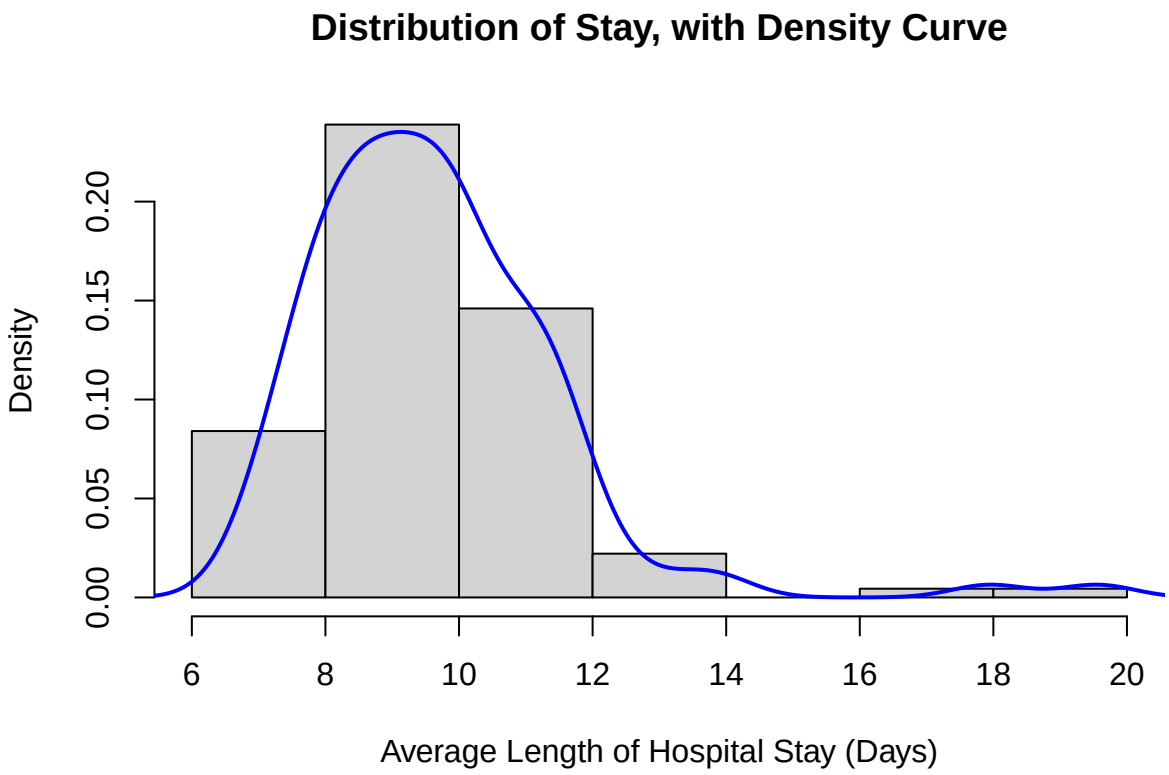


Figure 2: Histogram showing distribution of Stay Variable, with Density Curve Overlaid

Table 3: Age Descriptive Statistics

statistic	value
Min	38.8
Mean	53.23
Median	53.2
Max	65.9
StdDev	4.46

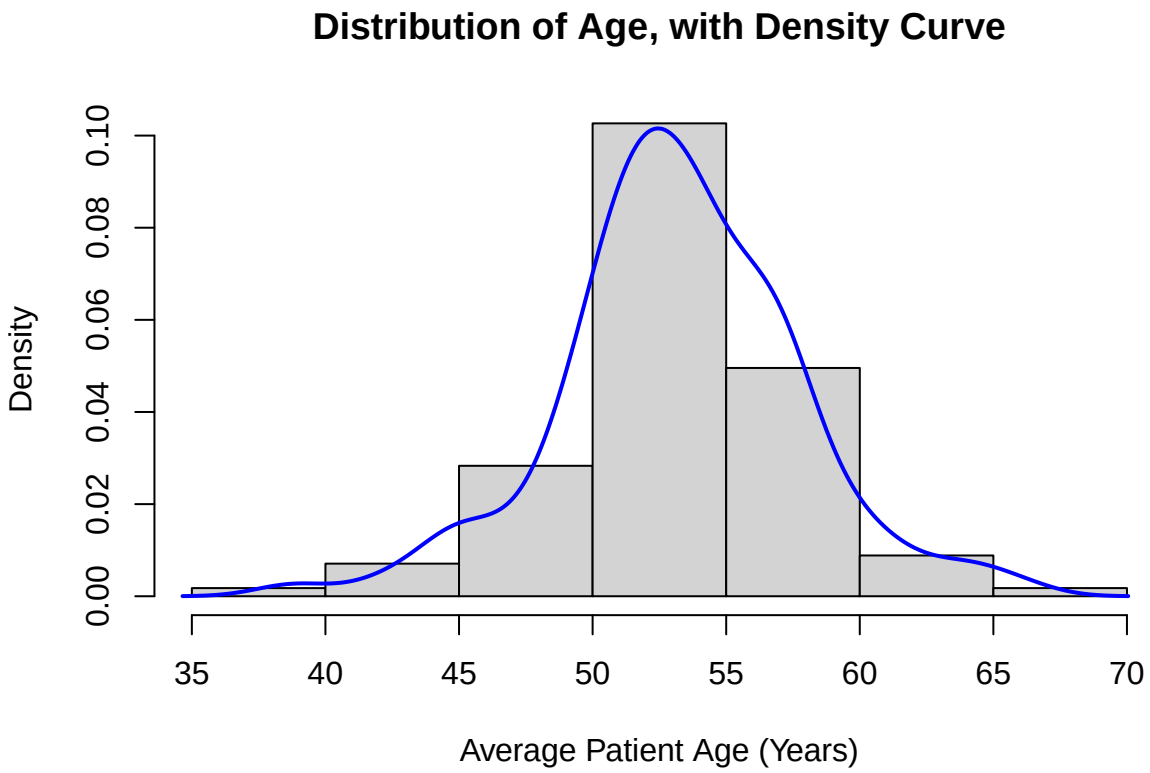


Figure 3: Histogram showing distribution of Age Variable, with Density Curve Overlaid

3.2 Hospital Factors

There are eight "Hospital" factors in our dataset: Culture, Xray, Beds, MedSchool, Region, Census, Nurses, and Facilities. We explore each variable below.

3.2.1 Culture

The variable Culture measures the ratio of number of cultures (testing a patient's sample for infection-causing microorganisms) performed to number of patients without signs or symptoms of hospital-acquired infection, times 100. For example, if we use the average ratio of number of cultures from this variable of 15.79, that

means that for every 100 hospital patients without signs or symptoms of hospital-acquired infection, 15.79 cultures were performed.

The histogram distribution is skewed to the right, with an average between 10 and 20. The descriptive statistics table "Culture Descriptive Statistics" prove this average to be actually 15.79. The minimum ratio of number of cultures is found to be 1.6 (a value fitting between 0 and 10 in the histogram), and the maximum ratio of number of cultures is 60.5 (a value fitting between 60 and 70 in the histogram).

The density curve fits the distribution pretty well. Additionally with the right-skewed distribution, there are outliers located between the bounds 50 and 70. We can interpret this as some hospitals having conducting a very high number of cultures on patients without signs or symptoms of hospital-acquired infection.

Table 4: Culture Descriptive Statistics

statistic	value
Min	1.6
Mean	15.79
Median	14.1
Max	60.5
StdDev	10.23

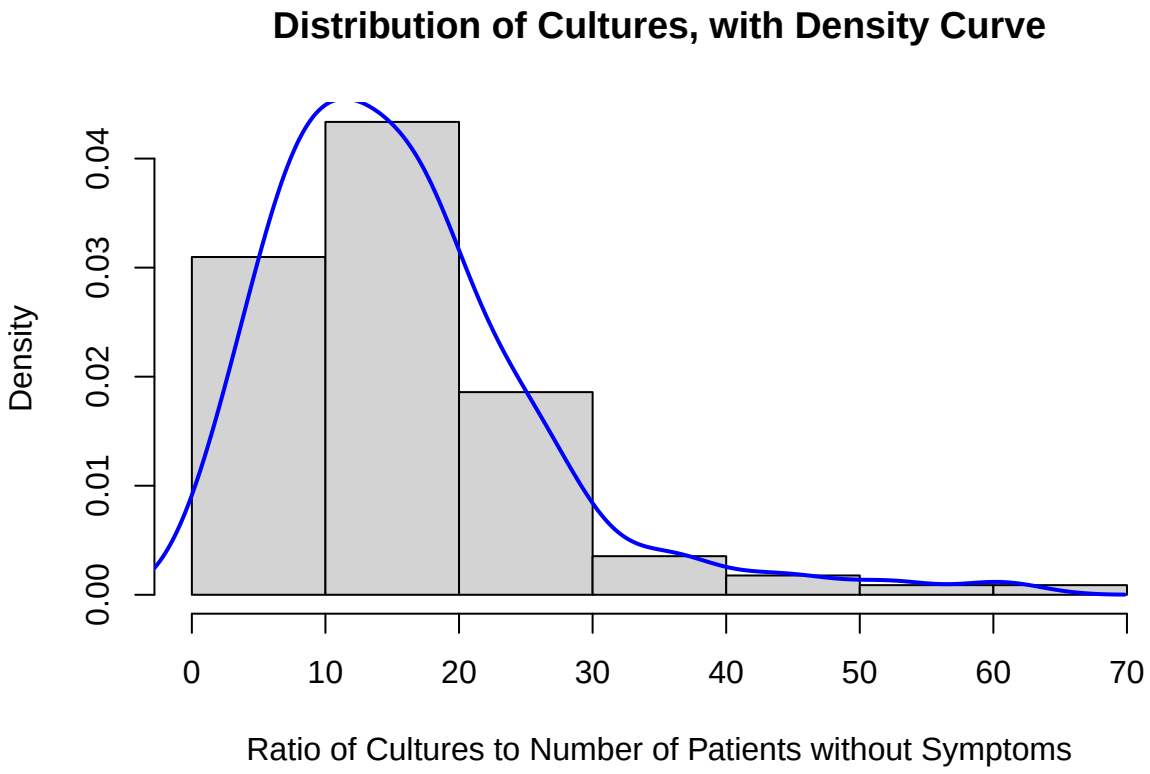


Figure 4: Histogram showing distribution of Culture Variable, with Density Curve Overlaid

3.2.2 Xray

The variable Xray measures the ratio of number of X-rays (taking an image of the body, particularly the chest, to look for conditions such as pneumonia) performed to number of patients without signs or symptoms of pneumonia, times 100.

The distribution of Xray is approximately bimodal as there are two distinct peaks: one between the bounds 80 and 90, and another between the bounds 50 and 60. This potentially indicates that there are two different groups in the dataset. For example, one group could be hospitals taking a lower number of Xrays for patients, and the other group could be taking a high number of Xrays on patients.

The descriptive statistics table "Xray Descriptive Statistics" reveals that the minimum ratio of number of X-rays performed to number of patients without symptons of pneumonia is 39.6, and the maximum ratio is 133.5. We find that the mean ratio is 81.63, and the median is 82.3.

The density curve fits the distribution pretty well.

Table 5: Xray Descriptive Statistics

statistic	value
Min	39.6
Mean	81.63
Median	82.3
Max	133.5
StdDev	19.36

3.2.3 Beds

The variable Beds measures the average number of beds in a hospital during study period.

The distribution of Beds from the histogram is skewed to the right, meaning there are outliers in the 600+ bed range. The minimum number of beds is 29 beds at one hospital, and the maximum number of beds is 835, which falls outside of our scale in the histogram. The mean number of beds is 252.17 beds, or about 252 beds, and the median number of beds is 186.

The density curve fits the distribution pretty well.

Table 6: Beds Descriptive Statistics

statistic	value
Min	29
Mean	252.17
Median	186
Max	835
StdDev	192.84

3.2.4 MedSchool

The variable MedSchool measures whether a hospital has an associated medical school, with 1 being Yes and 2 being No. We convert the values "1" and "2" into levels ("Yes" and "No") in section 2 "Data Description and Processing" for analysis.

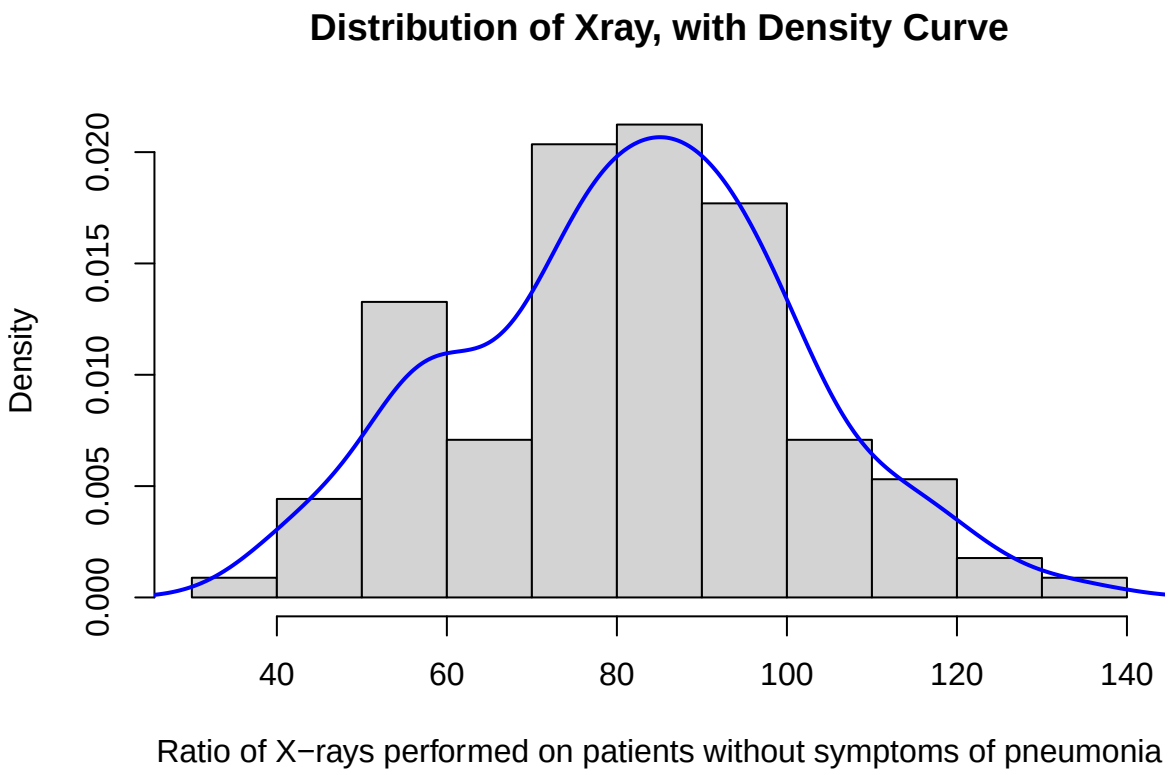


Figure 5: Histogram showing distribution of Xray Variable, with Density Curve Overlaid

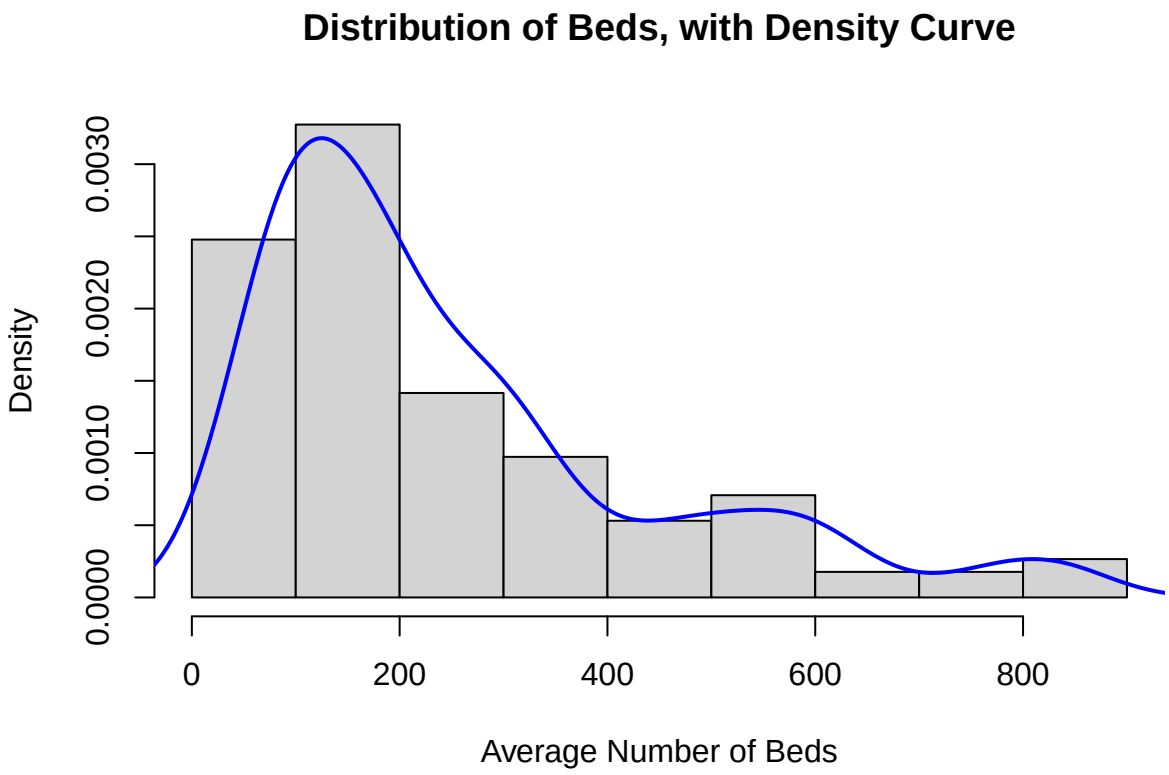


Figure 6: Histogram showing distribution of Beds Variable, with Density Curve Overlaid

From the table "Hospitals by Medical School Affiliation" and the bar chart "Bar Chart of MedSchool", we see that a majority of hospitals do not have an associated medical school, at 84.96%. There is a small minority of hospitals that have an associated medical school, at 15.04%.

The density curve fits the distribution pretty well.

Table 7: Hospitals by Medical School Affiliation

MedSchool	Count	Percent
Yes	17	15.04
No	96	84.96

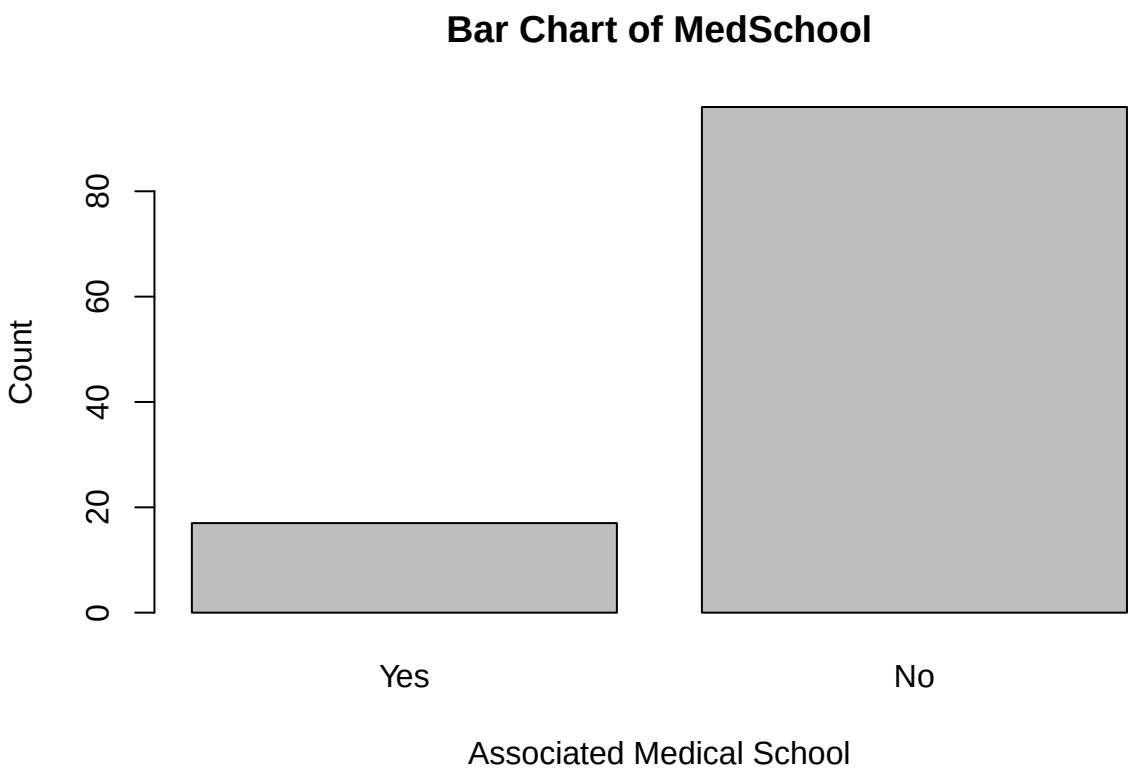


Figure 7: Bar Chart of the MedSchool Variable

3.2.5 Region

The variable Region measures the geographic region that a hospital is located in, where: 1 represents NE (Northeast), 2 represents NC (Northcentral), 3 represents S (South), and 4 represents W (West).

From the table "Hospitals by Geographic Region" and the bar chart "Bar Chart of Region", we see that most of the hospitals in our dataset are geographically located in the South, at 32.74%. Behind the South region is hospitals located in North Central, at 28.32%, followed by Northeast at 24.78%. The region with the least number of hospitals represented in our dataset is the West, at 14.16%.

The density curve fits the distribution pretty well.

Table 8: Hospitals by Geographic Region

Region	Count	Percent
Northeast	28	24.78
North Central	32	28.32
South	37	32.74
West	16	14.16

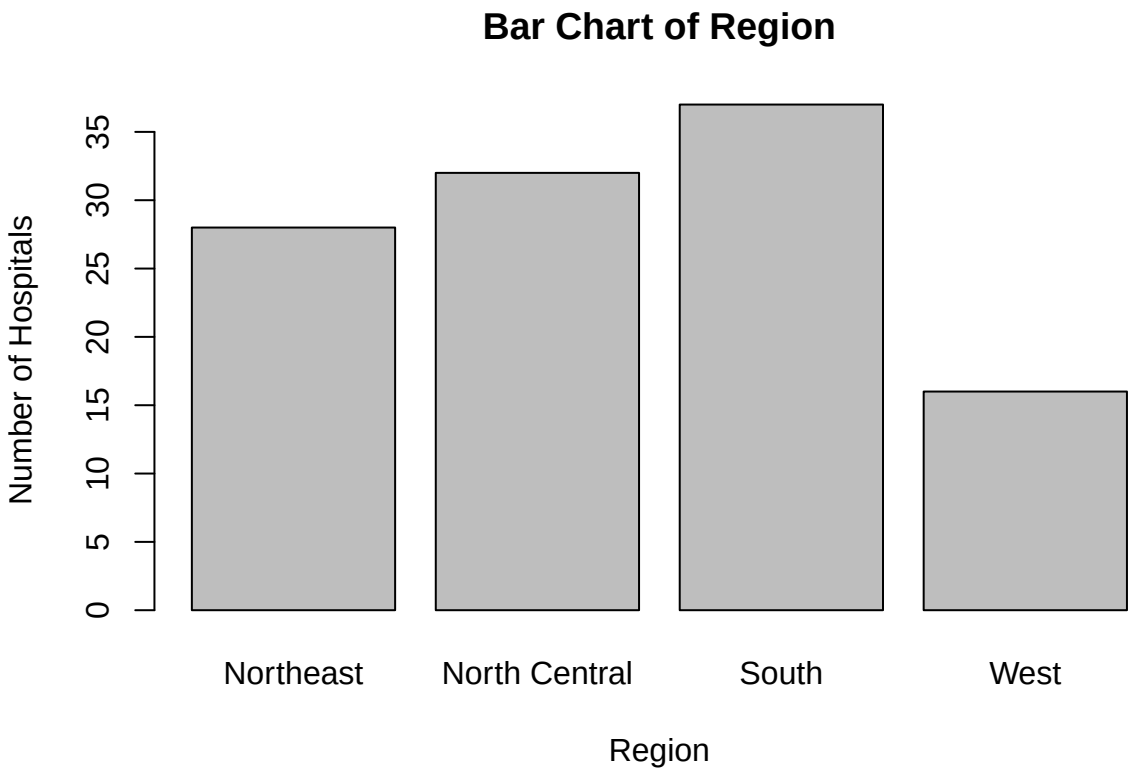


Figure 8: Bar Chart of the Region Variable, with Density Curve Overlaid

3.2.6 Census

The variable Census measures the average number of patients in hospital per day during the study period.

The distribution of Census is skewed to the right, with outliers between the bounds of 700 and 800 people in the histogram. The table "Census Descriptive Statistics" reveal that the minimum average number of people in a hospital is 20 people, the maximum average number of people in a hospital is 791, and the average number of people that visit a hospital is 191.37, or about 191 people. The median average number of people is 143.

The density curve fits the distribution pretty well.

Table 9: Census Descriptive Statistics

statistic	value
Min	20
Mean	191.37
Median	143
Max	791
StdDev	153.76

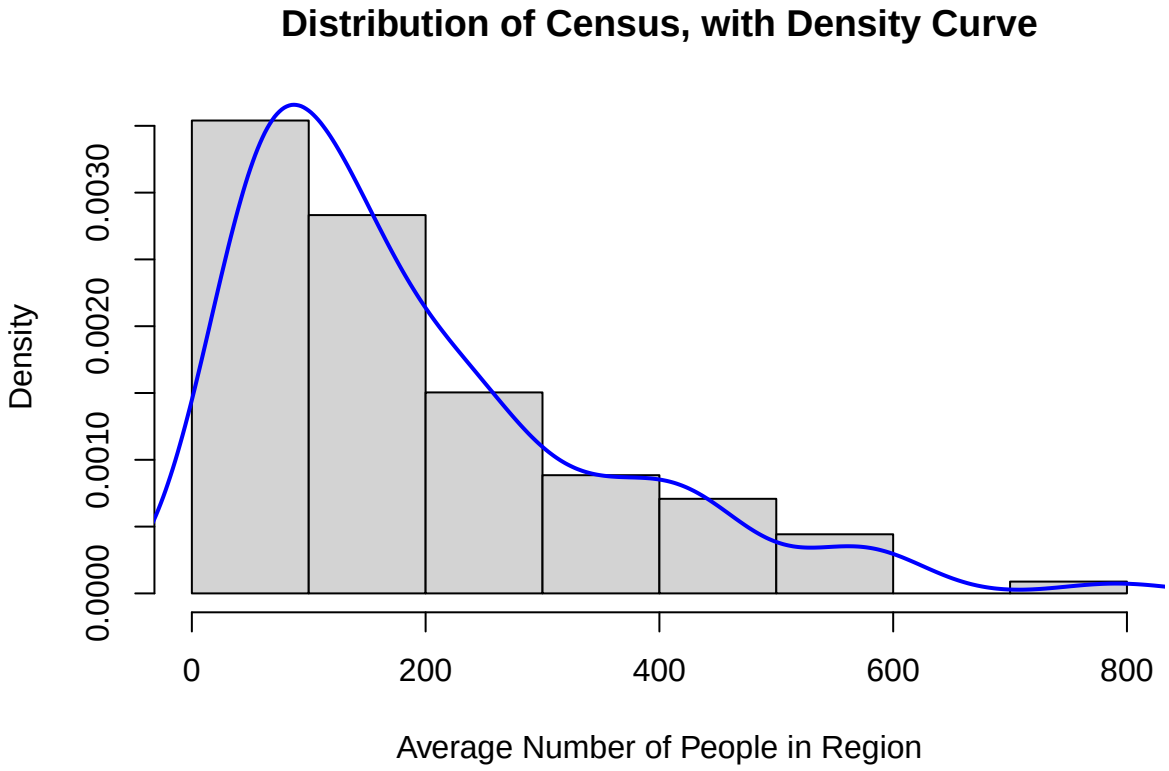


Figure 9: Histogram showing distribution of Census Variable, with Density Curve Overlaid

3.2.7 Nurses

The variable Nurses measures the average number of full-time equivalent registered and licensed practical nurses during study period. This number includes the number of full-time nurses plus one half the number of part-time nurses.

The distribution of Nurses is skewed to the right, with outliers between the bounds of 500 and 700 people in the histogram. A peak is found to be just below the 100 average number of nurses mark. The table "Nurses Descriptive Statistics" reveals that the minimum average number of nurses is 14, the maximum number is 656 nurses, and the mean average number of nurses is 173.25, or about 173 nurses. The median is 132 nurses. The right skewness of the histogram shows that there are few hospitals with an average number of Nurses over 500. Most of the hospitals have an average number of nurses in the range of 0 and 200, or about 173 based on the mean from the descriptive statistics.

The density curve fits the distribution pretty well.

Table 10: Nurses Descriptive Statistics

statistic	value
Min	14
Mean	173.25
Median	132
Max	656
StdDev	139.27

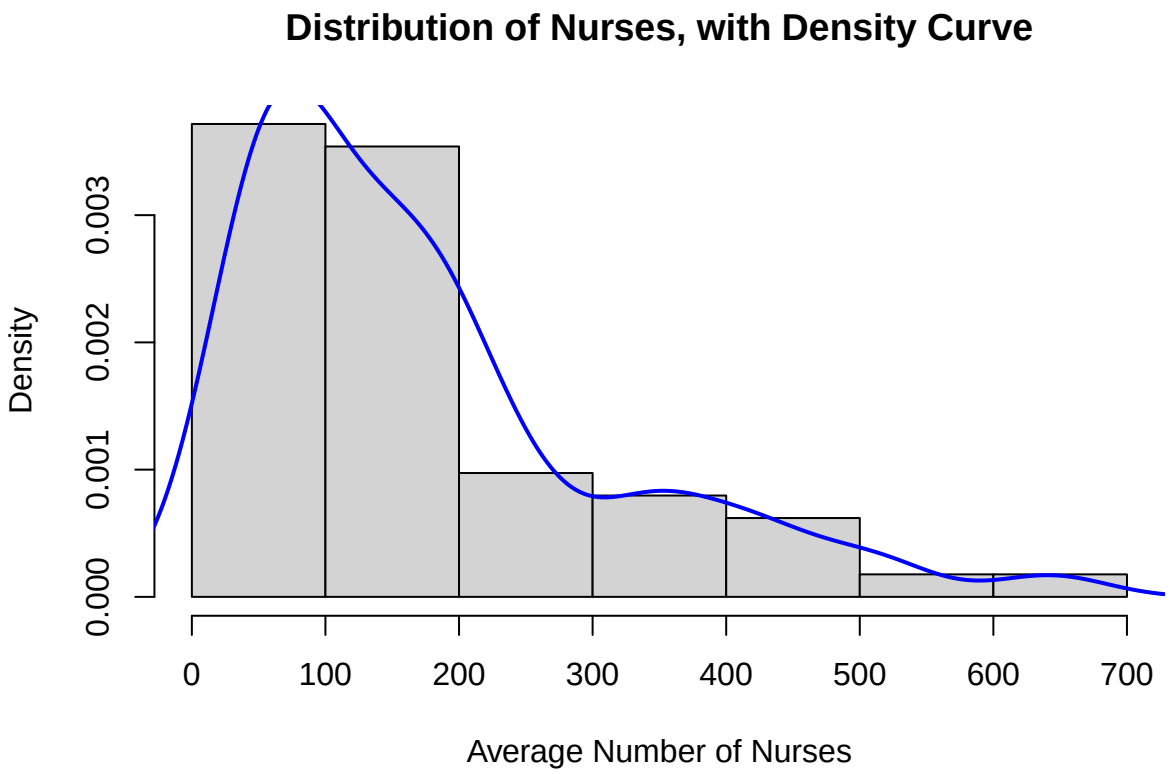


Figure 10: Histogram showing distribution of Nurses Variable, with Density Curve Overlaid

3.2.8 Facilities

The variable Facilities measures the percent of 35 potential facilities and services that are provided by the hospital.

The histogram of Facilities reveals a bimodal distribution, with two peaks: one peak between the bounds of 30 and 40, and another peak between the bounds of 50 and 60. This could mean that there could be two groups associated with this variable: a group of potentially rural hospitals with a lower percent of potential facilities and services, and another group of more wealthy/ middle class hospitals with a higher percent of 35 potential facilities and services at the hospital. The descriptive statistics reveal that the minimum percent

of 35 potential facilities and services that are provided by the hospital is 5.7%, and the maximum percent is 80%. The mean percent is 43.16%, and the median percent is 42.9%.

The density curve fits the distribution pretty well.

Table 11: Facilities Descriptive Statistics

statistic	value
Min	5.7
Mean	43.16
Median	42.9
Max	80
StdDev	15.2

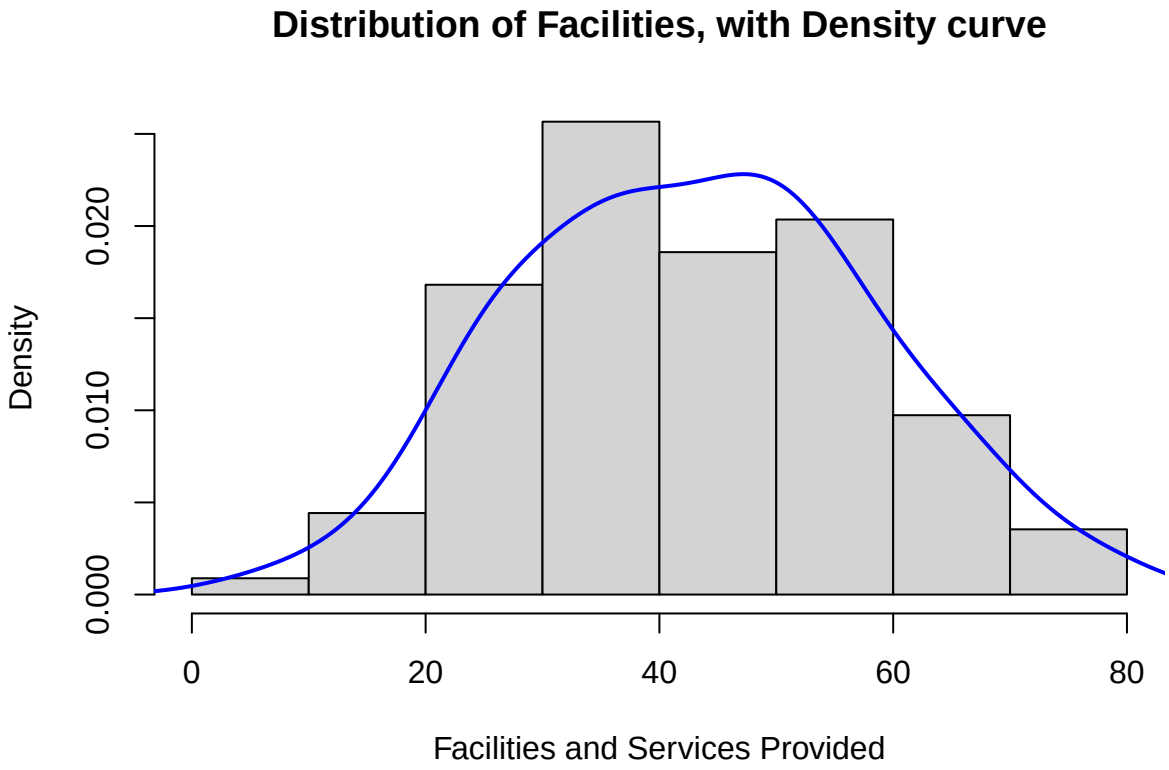


Figure 11: Histogram showing distribution of Facilities Variable, with Density Curve Overlaid

4 Modeling

For our models, infection risk is the dependent variable, and our ten predictors are: “Stay”, “Age”, “Culture”, “Xray”, “Beds”, “MedSchool”, “Region”, “Census”, “Nurses”, and “Facilities”.

4.1 Hypothesis

The null hypothesis is that all predictor coefficients, excluding the intercept, equal zero. This means the model provides no linear improvement over predicting infection risk using only its overall mean. The alternative hypothesis is that at least one predictor coefficient is nonzero.

Using a significance level of 0.05, the overall F-test was statistically significant ($F = 11.76$, $p < 0.01$). Therefore, we reject the null hypothesis and conclude that at least one predictor is associated with hospital infection risk.

4.2 Model 1:

For our first model, we will include all ten predictors.

Table 12: Regression Coefficients and Significance

Predictor	Coefficient	Standard Error	P-value
(Intercept)	-2.51	1.32	0.06
Stay	0.24	0.07	0.00
Age	0.01	0.02	0.54
Culture	0.05	0.01	0.00
Xray	0.01	0.01	0.03
Beds	0.00	0.00	0.20
MedSchoolNo	0.66	0.32	0.04
Region2	0.43	0.26	0.11
Region3	0.37	0.27	0.18
Region4	1.15	0.34	0.00
Census	0.00	0.00	0.27
Nurses	0.00	0.00	0.29
Facilities	0.02	0.01	0.04

Holding other predictors constant, Stay, Culture, Xray, Facilities, medical school affiliation, and the West to Northeast comparison were significant at the 0.05 level. Each additional day of average stay was associated with a 0.24 percent point increase in infection risk. Ten unit increases in Culture and Xray were associated with increases of 0.54 and 0.12 percentage points, respectively. A ten percent point increase in Facilities was associated with a 0.21 percentage-point increase.

Hospitals without medical school affiliations had infection risk 0.66 percentage points higher than affiliated hospitals. Western hospitals had risk 1.15 percentage points higher than Northeastern hospitals.

Age, Beds, Census, Nurses, and the remaining regional comparisons were not significant. Culture had the smallest p-value, providing the strongest evidence of an association. Note that coefficient magnitudes require caution when comparing predictors as their units differ.

Table 13: Overall Regression Model Fit

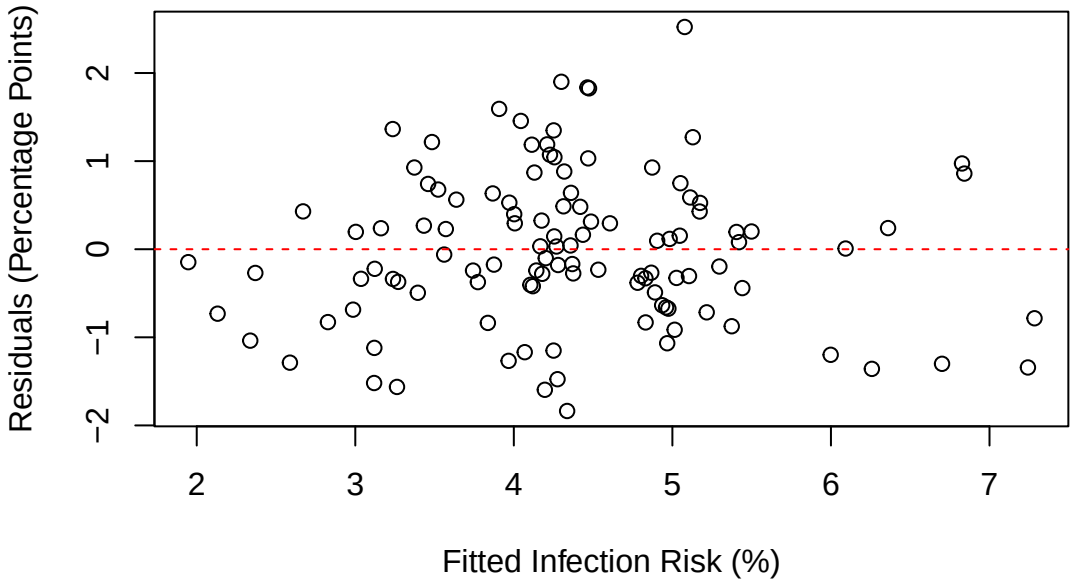
Measure	Value
R-squared (%)	58.54
Adjusted R-squared (%)	53.56
F-statistic	11.76
Model p-value	<0.01

The model explained 58.54% of the variation in infection risk across hospitals. After accounting for the predictor coefficients, the adjusted R-squared was 53.56%. The overall F-test was statistically significant ($F = 11.76$, $p < 0.01$); at least one predictor was associated with infection risk.

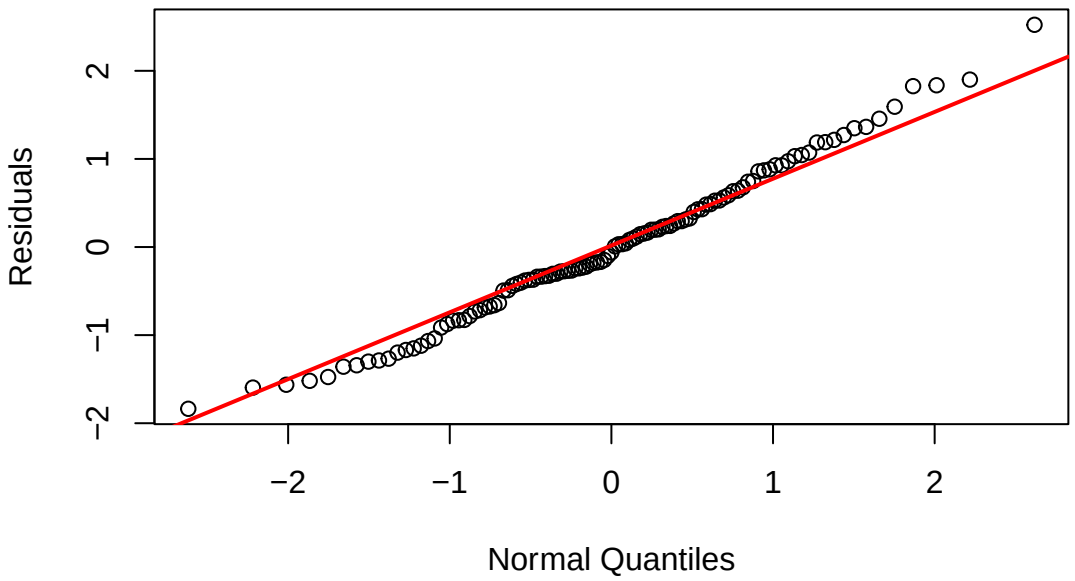
Table 14: Multicollinearity Assessment

Predictor	VIF
Stay	2.41
Age	1.27
Culture	1.56
Xray	1.39
Beds	35.74
MedSchool	1.79
Region	1.83
Census	37.77
Nurses	7.47
Facilities	3.14

The Multicollinearity assessment identified high VIF values for Beds (35.74), Census (37.77), and Nurses (7.47). These values indicate substantial overlap between these predictors and other predictors in the model. All remaining values were below the threshold(5).



The residuals are distributed on both sides of zero, without a pronounced funnel-shaped pattern. A few large residuals are also present. The plot therefore suggests possible departures from a linear relationship.



Most residuals follow the reference line closely. Departures occur in the tails, especially where the largest positive residuals lie above the line. This suggests residual normality, although the upper tail shows some deviation from the normal distribution.

Table 15: Breusch-Pagan Test for Nonconstant Variance

Test Statistic	Degrees of Freedom	P-value
14.01	12	0.3

The Breusch–Pagan test produced a test statistic of 14.01 with 12 degrees of freedom and a p-value of 0.30. At the 0.05 significance level, we fail to reject the null hypothesis of homoscedasticity (constant error variance). The test provides insufficient evidence of heteroscedasticity (nonconstant error variance), consistent with the absence of a funnel pattern in the residual plot.

4.3 Model 2:

For our second model, we will only include “Stay”, “Culture”, “Xray”, “MedSchool”, and “Region”, and “Facilities”. This model was chosen using stepwise selection.

Table 16: Regression Coefficients and Significance

Predictor	Coefficient	Standard Error	P-value
(Intercept)	-1.94	0.83	0.02
Stay	0.27	0.06	0.00
Culture	0.05	0.01	0.00
Xray	0.01	0.01	0.03
MedSchoolNo	0.50	0.29	0.09

Predictor	Coefficient	Standard Error	P-value
Region2	0.34	0.25	0.18
Region3	0.35	0.26	0.18
Region4	1.11	0.33	0.00
Facilities	0.03	0.01	0.00

Holding the other included predictors constant, Stay, Culture, Xray, Facilities, and the West to Northeast comparison were significant at the 0.05 level. Each additional day of average stay was associated with a 0.27 percent point increase in infection risk. One unit increases in Culture and Xray were associated with approximately 0.05 and 0.01 percent point increases, respectively. Each percentage-point increase in Facilities was associated with approximately 0.03 percent points higher infection risk.

Western hospitals had estimated infection risk 1.11 percent points higher than Northeastern hospitals. Medical school affiliation was not significant at the 0.05 level ($p = 0.09$), nor were the North Central to Northeast and South to Northeast comparisons.

Table 17: Overall Regression Model Fit

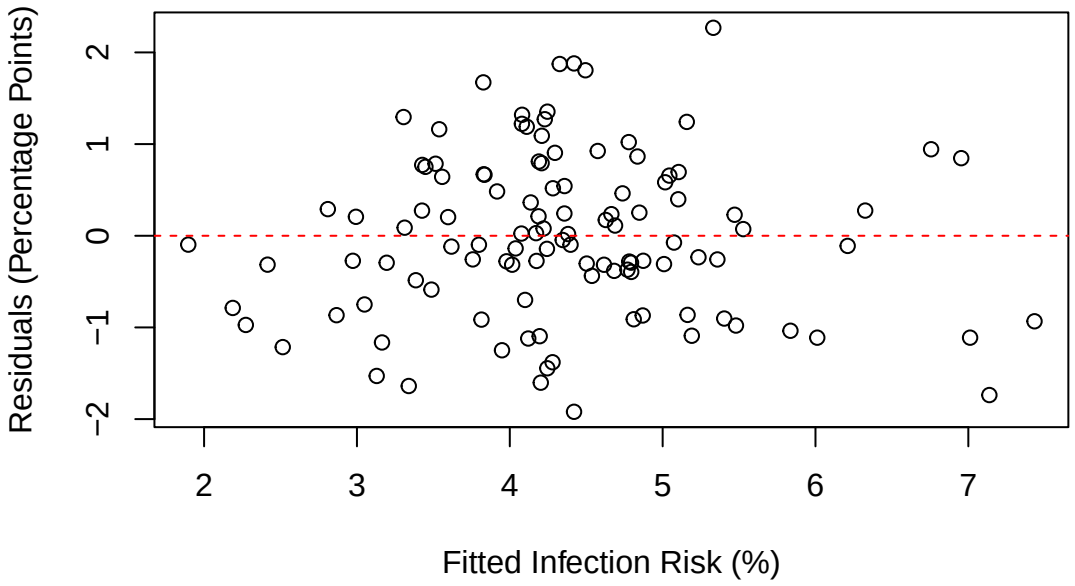
Measure	Value
R-squared (%)	57.12
Adjusted R-squared (%)	53.82
F-statistic	17.32
Model p-value	<0.01

Model 2 explained 57.12% of the variation in infection risk, with an adjusted R-squared of 53.82%. The overall F-test was statistically significant ($F = 17.32$, $p < 0.01$); at least one included predictor was associated with infection risk.

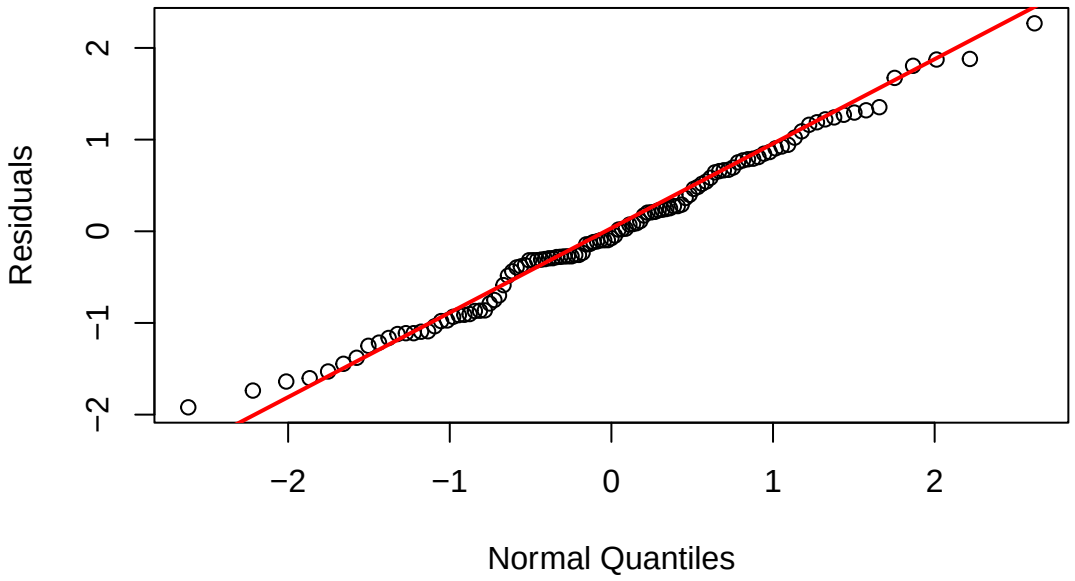
Table 18: Multicollinearity Assessment

Predictor	VIF
Stay	1.67
Culture	1.36
Xray	1.37
MedSchool	1.48
Region	1.50
Facilities	1.48

The displayed VIF values ranged from 1.36 to 1.67. Compared with Model 1, these results indicate reduced multicollinearity and less inflation of coefficient variances from overlapping predictors.



Residuals appear on both sides of zero. However, negative residuals at the lowest and highest fitted values suggest possible curvature. The reduced model therefore potentially departures from a linear relationship.



The residuals generally follow the normal reference line, with minor departures in the tails. The plot supports approximate residual normality.

Table 19: Breusch-Pagan Test for Nonconstant Variance

Test Statistic	Degrees of Freedom	P-value
6.83	8	0.56

The Breusch–Pagan test produced a statistic of 6.83 with 8 degrees of freedom and a p-value of 0.56. At the 0.05 significance level, we do not reject homoscedasticity (constant error variance). There is insufficient evidence of heteroscedasticity (nonconstant variance).

Model 2 retained six predictors rather than ten, while its adjusted R-squared increased slightly from 53.56% to 53.82%. It therefore achieved similar fit with fewer predictors and much less multicollinearity. # Predicting with Models 1 and 2

We evaluated both models using 10-fold cross-validation repeated three times. In each repetition, the data was divided into ten groups. Each model was trained using 90% of the observations and tested on the remaining 10%, until every group had been used for validation. Both models used the same splits, producing 30 validation results per model.

We measured prediction error using root mean squared error (RMSE), expressed in percentage points of infection risk. Lower RMSE indicates more accurate predictions. We summarized performance using the RMSE and its standard deviation across the validation samples.

Table 20: Repeated Cross-Validation Prediction Performance

Model	RMSE	RMSE Standard Deviation
Model 1: Full	0.99	0.19
Model 2: Reduced	0.95	0.16

Model 1 had a mean RMSE of 99 percentage points, compared with 95 for Model 2. Their RMSE standard deviations were 0.19 and 0.16, respectively. Model 2 had the lower average prediction error and showed more consistent performance across validation samples.

5 Summary and Conclusions